

University of Cape Town
Department of Electrical Engineering
Class Test 2
9 October 2025

EEE2047S: Signals and Systems I

Instructions

Time: 90 minutes Full marks = 60 marks

- Answer all questions.
- Write your answers on the question paper itself.
- Additional working may be written on the reverse sides (clearly indicate).
- Also provided: Information sheet with formulas.
- Write your EMPLID (PeopleSoft number) on the top right hand corner of this cover page.

Internal examiner: A/Prof A.J. Wilkinson

1. (12 marks) A voltage waveform is described by $v(t) = A \cos(\omega_0 t + \theta)$ with real constants $A > 0$, $\omega_0 > 0$ and $\theta = -\pi/2$.

- (a) Plot $v(t)$ and label significant points on the axes.
- (b) Determine the complex Fourier series and plot the magnitude and phase of the Fourier coefficients.
- (c) Determine the Fourier transform $V(\omega)$ and draw the spectrum.
- (d) Plot the instantaneous power as a function of time.
- (e) Derive the average power of $v(t)$ making use of Parseval's theorem.

2. (12 marks) A signal $x(t) = p_\tau(t)$ is fed into a LTI system with impulse response $h(t) = \text{Sa}(2\pi Bt)$. The parameters $\tau > 0$ and $B > 0$ are real constants. Let the output signal be $y(t)$.

- a) Plot $x(t)$
- b) Plot $h(t)$
- c) Find $Y(\omega) = \mathcal{F}\{y(t)\}$
- d) Plot $Y(\omega)$ for the case where $B = 2/\tau$ labelling significant features.

3. (10 marks) A signal $x(t)$ has Fourier transform $X(\omega)$.
Express the Fourier transforms of the functions below in terms of $X(\omega)$:

(a) $x_1(t) = x(1 - t) + x(-2 - t)$

(b) $x_2(t) = x(2t - 4)$

4. (5 marks) Use tables and properties to find

$$\mathcal{F}\left\{\frac{1}{5+jt}\right\}$$

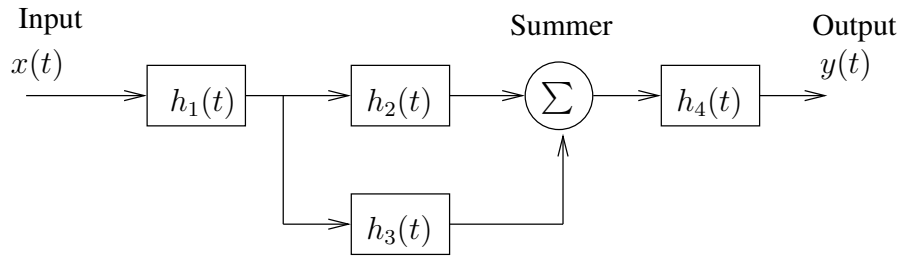
5. (13 marks) A system is described by the following differential equation

$$\frac{d}{dt}y(t) + y(t) = \frac{d}{dt}x(t) - x(t).$$

- (a) Derive the transfer function $H(\omega)$
- (b) Determine and plot the impulse response $h(t)$
- (c) Find the step response.

6. (8 marks) Block diagrams:

(a) Derive the system transfer function $H(\omega)$ in terms of the parameters given.



(b) Derive the transfer function $H(\omega)$ for the following feedback system:

